

# Problem Set 4

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## Question 1

We are interested in how income and occupational type are related to occupational prestige. We focus on whether professional occupations have higher prestige and whether the effect of income differs between professional and non-professional jobs.

### 1. Create the professional dummy variable

We create a dummy variable that equals 1 for professional occupations and 0 for blue- and white-collar occupations.

```
1  # Load data
2  library(car)
3  data(Prestige)
4
5  # Create dummy variable: professional = 1 if type == "prof",
6  # else 0
7  Prestige$professional <- ifelse(Prestige$type == "prof", 1, 0)
8
9  # Check coding
10 table(Prestige$professional, Prestige$type)
```

### 2. Regress prestige on income, professional, and interaction

We estimate the following model:

$$prestige_i = \beta_0 + \beta_1 income_i + \beta_2 professional_i + \beta_3 (income_i \cdot professional_i) + \varepsilon_i.$$

The regression is run in R as follows:

```
1  # Run regression with interaction
2  m1 <- lm(prestige ~ income * professional, data = Prestige)
3  summary(m1)
```

Table 1 reports the regression results.

Table 1: Outcome variable is prestige. Predictors are income, professional, and their interaction.

|                     | Model 1                      |
|---------------------|------------------------------|
| (Intercept)         | 21.1423***<br>(2.8044)       |
| income              | 0.0031709***<br>(0.0004993)  |
| professional        | 37.7813***<br>(4.2483)       |
| income:professional | -0.0023257***<br>(0.0005675) |
| R-squared           | 0.7872                       |
| Adj. R-squared      | 0.7804                       |
| Num. obs.           | 98                           |

Standard errors in parentheses. \*\*\*p<0.001, \*\*p<0.01, \*p<0.05.

### 3. Prediction equation

Using the estimated coefficients from Table 1, the fitted equation is

$$\widehat{prestige} = 21.1423 + 0.0031709 \cdot income + 37.7813 \cdot professional - 0.0023257 (income \cdot professional).$$

### 4. Interpretation of income coefficient

The coefficient on income is 0.0031709. This represents the marginal effect of income on prestige for non-professional occupations (professional = 0). Holding professional status fixed at 0, a one-unit (one dollar) increase in income is associated with an average increase of 0.00317 points in prestige. A 1,000-unit increase in income corresponds to an increase of about 3.17 points in prestige.

### 5. Interpretation of professional coefficient

The coefficient on the professional dummy is 37.7813. Because the model includes an interaction between income and professional, this coefficient captures the difference in predicted prestige between professional and non-professional occupations when income equals 0. At income = 0, professional occupations are predicted to have about 37.78 more prestige points than non-professional occupations.

## 6. Effect of a 1,000-unit income increase for professionals

With an interaction term, the marginal effect of income is

$$\frac{\partial \widehat{prestige}}{\partial income} = \beta_1 + \beta_3 \cdot professional.$$

For professional occupations ( $professional = 1$ ), this becomes

$$ME_{income|prof=1} = 0.0031709 - 0.0023257 = 0.0008452.$$

Thus, for professionals, a 1,000-unit increase in income changes prestige by

$$1000 \times 0.0008452 \approx 0.8452.$$

The corresponding R code is:

```
1  beta <- coef(m1)
2  marg_income_prof1_per1 <- beta["income"] + beta["income:
   professional"]
3  marg_income_prof1_1000 <- marg_income_prof1_per1 * 1000
```

This shows that income has a positive but comparatively small effect on prestige among professional occupations.

## 7. Effect of switching to professional at income = 6000

The effect of changing from non-professional ( $professional = 0$ ) to professional ( $professional = 1$ ) at a given income level is

$$\Delta prestige = \beta_2 + \beta_3 \cdot income.$$

At income = 6000, we obtain

$$\Delta prestige = 37.7813 + 6000(-0.0023257) = 23.8270.$$

In other words, at an income level of 6,000, professional occupations are predicted to have about 23.83 more prestige points than non-professional occupations.

The calculation is implemented in R as:

```
1  effect_prof_switch_6000 <-
2  beta["professional"] + 6000 * beta["income:professional"]
```

## Question 2

We now consider an experiment on campaign yard signs in a gubernatorial election. The outcome variable is the vote share of Cuccinelli, and the key predictors are indicators for whether a precinct was assigned yard signs and whether it was adjacent to precincts with yard signs. The regression model is

$$voteshare_i = \beta_0 + \beta_1 assigned_i + \beta_2 adjacent_i + \varepsilon_i.$$

The reported coefficients are summarized in Table 2.

Table 2: Effect of campaign yard signs on Cuccinelli's vote share.

|                        | Model 1             |
|------------------------|---------------------|
| (Intercept)            | 0.302***<br>(0.011) |
| Assigned lawn signs    | 0.042**<br>(0.016)  |
| Adjacent to lawn signs | 0.042***<br>(0.013) |
| R-squared              | 0.094               |
| Num. obs.              | 131                 |

Standard errors in parentheses. \*\*\*p<0.01, \*\*p<0.05.

## 1. Do yard signs affect vote share?

We first test whether having yard signs in a precinct affects vote share. The null and alternative hypotheses for the coefficient on assigned are:

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0.$$

The estimated coefficient is 0.042 with a standard error of 0.016. The t-statistic is

$$t = \frac{0.042}{0.016} \approx 2.63.$$

Using a two-sided test with significance level 0.05, the critical value is about 1.98. Since the absolute t-statistic is larger than 1.98, we reject the null hypothesis and conclude that yard signs have a statistically significant effect on vote share. Substantively, precincts that were assigned yard signs are predicted to give Cuccinelli about 4.2 percentage points more of the vote than similar precincts without signs.

## 2. Do adjacent precincts show an effect?

We next consider the effect of being adjacent to precincts with yard signs. The estimated coefficient is again 0.042, with a standard error of 0.013. The t-statistic is

$$t = \frac{0.042}{0.013} \approx 3.23.$$

This value is also larger than 1.98 in absolute value, so we reject the null hypothesis that the coefficient on adjacent is zero. We conclude that precincts adjacent to yard signs also show a statistically significant increase in Cuccinelli's vote share of about 4.2 percentage points.

### **3. Interpretation of the intercept**

The intercept is estimated as 0.302. This represents the expected vote share for Cuccinelli in precincts that neither had yard signs nor were adjacent to precincts with yard signs. In the absence of any yard sign exposure, Cuccinelli is predicted to receive about 30.2 percent of the vote.

### **4. Model fit and importance of yard signs**

The model has R-squared equal to 0.094, meaning that the two yard sign variables explain about 9.4 percent of the variation in vote share across precincts. This indicates that while yard signs and adjacency have statistically detectable effects, most of the variation in vote share is driven by other factors not included in the model, such as demographics, partisanship, candidate quality, or other campaign activities. Yard signs matter, but they are only one of many determinants of electoral outcomes.